| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 6(a) | $\mathrm{G}_X'(t)=\dfrac{(3-2t)^2+4(3-2t)t}{(3-2t)^4}\left[=\dfrac{3+2t}{(3-2t)^3}\right]$ | **M1** | Attempt to differentiate $\mathrm{G}_X(t)$.<br>May be implied by expressions for $\mathrm{G}_X'(1)$. |
| | $\mathrm{E}(X)=\mathrm{G}_X'(1)=5$ | **A1** | |
| | $\mathrm{G}_X''(t)=\dfrac{2(3-2t)^3+6(3+2t)(3-2t)^2}{(3-2t)^6}\left[=\dfrac{24+8t}{(3-2t)^4}\right]$ | **M1** | Attempt to differentiate $\mathrm{G}_X'(t)$.<br>May be implied by expressions for $\mathrm{G}_X''(1)$. |
| | $\mathrm{Var}(X)=\mathrm{G}_X''(1)+\mathrm{G}_X'(1)-\left(\mathrm{G}_X'(1)\right)^2=32+5-25$ | **M1** | Use correct formula using *their* values. |
| | $\left[\mathrm{Var}(X)=\right]12$ | **A1** | CWO |
| | | **5** | |

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|---|---|---|---|
| 6(b) | $G_Z(t) = \frac{t^3}{(3-2t)^4} = \frac{1}{81}t^3\left(1+(-4)\left(-\frac{2}{3}t\right)+\ldots\right)$ | M1 | Multiply and attempt at binomial expansion. |
| | $P(Z=4)$ = *their* coefficient of $t^4$.<br>OR $P(Z=3)$ = *their* coefficient of $t^3$. | M1 | Attempt at finding *their* coefficient of $t^3$ or $t^4$.<br>Either seen earns M1. |
| | $\left[P(Z=4)=\right]\frac{8}{243}$ and $\left[P(Z=3)=\right]\frac{1}{81}$ | A1 | Both values seen. |
| | $P(Z>4) = 1-{'}\frac{8}{243}{'}-{'}\frac{1}{81}{'}$ | M1 | Attempt to find $P(Z>4)$ using *their* values. |
| | $P(Z>4)=\frac{232}{243}$ | A1 | AWRT 0.955.<br>Note: considering the expansions of the probability generating functions for $X$ and $Y$ separately, then finding $P(Z>4)=1-P(Z\leqslant 4)$ with $P(Z\leqslant 4)=P(X=1,Y=2)+P(X=1,Y=3)+P(X=2,Y=2)$ earns full credit. |
| | | 5 | |